

Ayush Kumar

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EDUCATION

- Anna University** Chennai, India
• *B.E - Computer Science and Engineering; GPA: 8.5* July 2022 - June 2026

SKILLS SUMMARY

- **Languages:** Python, MySQL
- **Frameworks:** NumPy, Pandas, Matplotlib, Seaborn, Gen AI, LangChain, LangGraph, Scikit-learn, TensorFlow, Keras, Flask
- **Tools:** VS Code, Jupyter, Anaconda Navigator, Power BI, Tableau, Excel
- **Cloud:** AWS (Basics), Azure (Basics)
- **Version Control:** Git, GitHub
- **Machine Learning:** Supervised & Unsupervised Learning, Regression & Classification, Feature Engineering, Model Evaluation, Data Preprocessing & Cleaning, EDA, CNN, Computer Vision, ANN, RNN

EXPERIENCE

- **Data Science Intern – ScoreSight: Predicting EPL Points Tally — GitHub** Remote
Infosys Aug 2025 - Oct 2025
 - Built an EPL match prediction system using ensemble ML (Random Forest + XGBoost + Logistic Regression) achieving 75.7% pre-match & 68.1% half-time accuracy.
 - Developed a full-stack app (FastAPI + React + TypeScript) with JWT auth, processing 10+ years of EPL data to extract 15+ features for model training.
 - Integrated an AI football chatbot with real-time analytics dashboard tracking team form, head-to-head stats & strength scores for instant insights.
- **Data Analytics Intern – Visualizing US Natural Disaster Declarations — GitHub** Remote
Infosys Nov 2025 - Mar 2026
 - Built 3 Power BI dashboards analyzing 6+ KPIs from FEMA disaster data across 50 US states, identifying top hotspot regions and rising disaster trends.
 - Cleaned and transformed large-scale FEMA dataset using Power Query & Python (Pandas) covering disaster declarations, incident types and 4 assistance programs.
 - Uncovered floods, hurricanes & severe storms as top disaster types across 5 high-impact states with prolonged recovery durations via EDA and trend analysis.

PROJECTS

- **Real-Time Depression Detection via ResNet-50 — GitHub**
Python, ResNet-50, OpenCV, CNN, Image Processing, Feature Engineering Jan 2026 - Jun 2026
 - Built a real-time depression detection system using ResNet-50 & CNN trained on FER2013 dataset, achieving 72% training accuracy & 68% validation accuracy across 7 facial emotion classes.
 - Designed end-to-end deep learning pipeline covering image preprocessing, grayscale conversion, feature extraction and emotion classification across 48x48 facial images with Dropout regularization over 10 epochs.
 - Developed Flask web app with real-time stress level estimation & emotion-based song recommendation system, enabling non-invasive automated mental health screening support.
- **Wafer Fault Detection — GitHub**
Python, Scikit-learn, Flask, SMOTE, XGBoost, Random Forest Mar 2025 - May 2025
 - Built a Wafer Fault Detection system using Balanced Random Forest on 590 sensor features, achieving 92% accuracy with SMOTE to handle highly imbalanced semiconductor dataset.
 - Engineered end-to-end ML pipeline with missing value imputation, Variance Threshold feature selection, and evaluated 4 models (Logistic Regression, Random Forest, XGBoost, Balanced Random Forest).
 - Deployed Flask web app enabling real-time CSV upload, automated preprocessing and Good/Bad wafer prediction using saved Joblib model.
- **Cement Strength Prediction — GitHub**
Python, Scikit-learn, Flask, Pandas, Joblib, EDA, Feature Engineering, Regression Modeling Jun 2025 - Aug 2025
 - Predicted cement compressive strength using Random Forest Regressor, achieving R^2 of 0.91 vs Linear Regression R^2 of 0.58 across 8 concrete mixture features.
 - Applied EDA and feature engineering on 8 components (cement, fly ash, water, age) to identify key strength predictors in MPa.
 - Deployed real-time Flask web app with Joblib serialization for instant compressive strength prediction from mixture inputs.

CERTIFICATIONS

- IIRS-ISRO Online Course: AI/ML for Geodata Analysis – May, 2024
- Microsoft Certified: Power BI Data Analyst Associate (PL-300) – Microsoft, 2025
- iNeuron.ai Data Science Professional Certificate – August, 2024